UNIT- 1

TITLE OF THE UNIT- FUELS

Abstract

Energy is necessary to do work. Energy is produced when fuel undergoes combustion. Thermochemistry plays a vital role in understanding the energy transformations in fuels, particularly through the calorific value, which measures energy released during combustion. This unit deals with different types of calorific values, knocking in petrol and diesel engine, knocking in terms of octane no and cetane no and solutions to overcome adverse effect of knocking. As it is known after few years coal and petroleum will not be available, so we will be discussing some inexhaustible source of energy. A special category of fuel and oxidizer is required to undergo combustion to give high thrust to rocket. Briefly we will discuss about the rocket propellant types, characteristics and properties.

Alternative Fuels: Green fuels like hydrogen and biodiesel contribute to sustainable energy goals by reducing greenhouse gas emissions. Advanced technologies in fuel synthesis and storage, along with innovations in rocket propulsion systems, continue to drive progress in achieving efficient and environmentally friendly energy solutions across industries.

Blow-up Syllabus

Sl No	Topic to be taught	Duration
1	Introduction to Thermochemistry, calorific value of fuels	1 hour
2	Numericals on calorific values, knocking in internal combustion engines	1 hour
3	Reasons for knocking in IC engine, Knocking in disel engine	1 hour
4	Octane and Cetane number	1 hour
5	Anti knocking agents. Biodiesel-synthesis, power alcohol	1 hour
6	Green fuel- hydrogen production and storage	1 hour
7	Rockets Fuels: Properties	1 hour
8	Characteristics of Rocket Propellant and types	1 hour

UNIT- I

Thermochemistry

Thermochemistry deals with the study of heats of chemical reactions. When a chemical reaction takes place, energy in various forms may either be emitted or absorbed. During many reactions the temperature rises indicating that heat is being evolved. In others, when temperature falls, absorption of heat is indicated. Thus, the study of the heat produced or required by chemical reaction forms the basis of thermochemistry. The reactions in which heat escapes from the system to the surroundings are termed as exothermic processes that means the surrounding ultimately becomes hotter. The release of heat leads to decrease in the enthalpy of a system when the pressure remains constant, thus the enthalpy change is negative. The chemical processes in which heat is absorbed by the system from the surroundings are endothermic processes that means the surroundings ultimately become colder. Here the absorption of heat leads to increase in the enthalpy of a system at constant pressure, thus the enthalpy change is positive.

A fuel is defined as naturally occurring or artificially manufactured combustible carbonaceous material which serves particularly as source of heat and light and also in few cases as a source of raw material.

Fuel: A fuel is the substance which on combustion produces a large amount of heat.

$$\text{Fuel} + O_2 \xrightarrow{\text{high temp}} \text{Combustion products} + \text{Heat}$$

As most of the fuels contain carbon or carbon and hydrogen, the combustion involves the oxidation of carbon-to-carbon dioxide and hydrogen to water. Sulphur, if present, is oxidized to sulphur dioxide while the mineral matter forms the ash.

Importance of hydrocarbons as fuels:

Fossil fuels contain mainly hydrocarbons. These hydrocarbons are important sources of energy in daily life. Hydrocarbons are used as energy sources in cooking, lighting, automobiles, production of electricity in thermal power plants etc. These hydrocarbon fuels meet 80% of the world's energy demand. Thus, hydrocarbons are important sources of energy.

Characteristics of a Good/ideal fuel:

While selecting an ideal fuel for domestic or industrial purpose we should keep in mind that The fuel selected must possess the following characteristic properties.

1. It should possess high calorific value.
2. It should have a proper ignition temperature. The ignition temperature of the fuel should neither be too low nor too high.

A high calorific value refers to the large amount of heat energy produced when a unit mass or volume of a substance,

3. It should have a proper ignition temperature. The ignition temperature of the fuel should neither be too low nor too high.
4. It should not produce poisonous products during combustion. In other words, it should not cause pollution on combustion.
5. It should have a moderate rate of combustion.
6. Combustion should be easily controllable i.e., combustion of fuel should be easy to start or stop as and when required.
7. It should not leave behind much ash on combustion.
8. It should be easily available in plenty and cheap.
9. It should have low moisture content.

Classification of fuels: Fuels are classified into two types.

Based on their origin they are classified into

- Primary fuels
- Secondary fuels

Primary Fuels: There are naturally occurring fuels which serves as source of energy without any chemical processing.

Ex: Wood, Coal, Crude oil, Natural gas, Peat, Lignite, Anthracite.

Secondary Fuels: -They are derived from primary fuels & serves as source of energy only after subjecting to chemical processing.

Ex: Charcoal, Coke, producer gas, Petrol, Diesel etc.,

Based on their physical state fuel are classified into

- Solid
- Liquid
- Gaseous fuels

Fuels					
Primary			**Secondary**		
Solid	**Liquid**	**Gas**	**Solid**	**Liquid**	**Gas**
Coal, Wood	Crude Oil (petroleum)	Natural gas	Charcoal, coke	Petrol, Diesel	Coal gas, water gas

Table 1.1: Fuel classification

Calorific value

Calorific value is defined as the amount of heat liberated when a unit mass of fuel is burnt completely in presence of air or oxygen.

Calorific value is of two types as follows:

Higher calorific value. (HCV) or Gross calorific value. (GCV)

Lower calorific value. (LCV) or Net calorific value. (NCV)

HCV: - It is the amount of heat liberated when a unit mass of fuels burnt completely in the presence of air or oxygen and the products of combustion are cooled to room temperature. Here it includes the heat liberated during combustion and the latent heat of steam. Hence its value is always higher than lower calorific value.

LCV: - It is the amount of heat liberated when a unit mass of fuel is burnt completely in the presence of air or oxygen and the product of combustion is let off completely into air. It does not include the latent heat of steam. **Therefore, it is always less than HCV.**

Calculation of Heat associated with steam

Let the percentage of hydrogen in the fuel be "H"

2 kg of hydrogen produces 18 kg of water

Therefore, H kg of Hydrogen produces $(\frac{18}{2})$ x H kg of water

Water produced by 1 kg of fuel = $(\frac{18}{2})$ x (H/100) $(\frac{H}{100})$ kg = 0.09 H kg

Heat associated with steam = weight of water produced x Latent heat of condensation or vaporization

= 0.09H x 587 x 4.2 kJ/kg

(18 is the molecular weight of H_2O and 587 is the latent heat of condensation in calorie per gram)

∴ Gross calorific value = Net Calorific value + Latent Heat of condensation or vaporization of water produced.

Numericals

(1) Find out the percentage of Hydrogen in the 5g of Hydrocarbon fuel burnt in air and heat carried away by H2O is 250kJ/kg

Ans: Heat associated with steam = 0.09H x 587 x 4.2 kJ/kg

H = (Heat associated with steam)/ (587 x 4.2 x 0.09) = 1.126

(2) 3.1kg of fuel was burnt in air and combustion products were allowed to cool at 298 K. The calorific value obtained by the process is 500 kJ/kg. If the % of hydrogen in the fuel is 2 then what is the net calorific value.

Ans: % of hydrogen = 2

100 kg fuel contains 2 kg of hydrogen

GCV = 500 kJ/kg

NCV = (GCV – (0.09H x 587 x 4.2)) kJ/kg

= (500 – (0.09*2*587*4.2)) kJ/kg

= 56.22 kJ/kg

Knocking of IC Engine:

Knocking in Petrol Engines:

In petrol engines, the mixture of petrol and air is drawn in to the cylinder. The fuel- air mixture is compressed by the piston and is ignited by an electric spark. As the flame front travels in the combustion chamber, rapidly expanding combustion products compress the remaining unburnt fuel and raise its temperature. If the flame front travels rapidly at an optimum speed, the combustion of unburnt fuel takes place smoothly. On the other hand, if the flame front travels too slowly, the entire last portion of fuel mixture may get heated up beyond its ignition temperature and undergo instantaneous explosive combustion. This result in emission of a characteristic rattling sound called "knocking".

Knocking is the rattling noise or metallic sound produced in an internal combustion engine due to uneven combustion (detonation) of fuel and air mixture.

The reasons for Knocking are

(1) Very high compression ratio causes pre-ignition.

Compression ratio is the ratio of the volume of the fuel and air mixture (V1) at the end of suction stroke to the volume of the mixture (V2) at the end of compression stroke.

Compression ratio (CR) = $\frac{V1}{V2}$

The value (CR) will always be greater. Power output of the IC engine increases continuously with increase in compression ratio. The compression ratio corresponding to maximum power output is known as the highest useful compression ratio (HUCR).Very high compression ratio means that the fuel and air are compressed to a maximum extent, leads to an increase in the temperature of the cylinder. Hot spots are formed inside. When the fuel and air mixture are

compressed, hot spots ignite the fuel even before the spark plug produces spark. This phenomenon is called pre ignition which leads to knocking.

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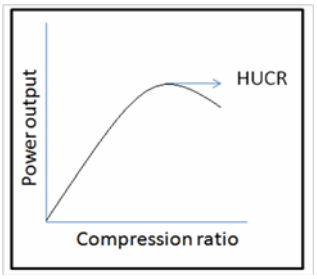


Figure 1.1: Highest useful compression ratio

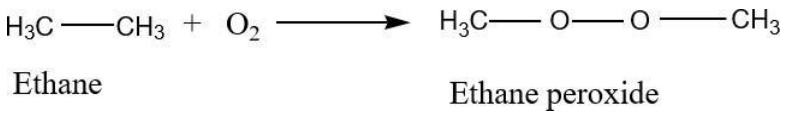
(2) Peroxide formation leads to self-Ignition

A. **Under Normal** Conditions there is a slow oxidation of the fuel.

Normal combustion takes place due to the chain reaction. The overall reaction may be represented as $C_2H_6 + \frac{3}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

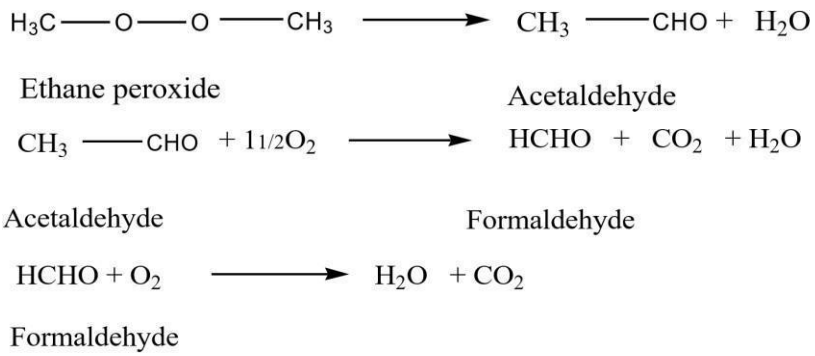
B. **Under Knocking Conditions**, the rate of combustion is very high. It involves following steps.

(i) Oxygen combines with hydrocarbon molecules forming peroxides.



Ethane — Ethane peroxide

The peroxides decompose readily to give a number of gaseous products.
For example,



Ethane peroxide — Acetaldehyde

Acetaldehyde — Formaldehyde

Formaldehyde

Fast reaction leads to rapid increase of pressure. This **results in knocking.**

(3) Chemical Structure of the Fuel:

Knocking tendency decreases with the increase in compactness of the molecule and increases with the increase in the length of hydrocarbon chain.

Ex., i) 2, 2-dimethyl pentane knocks lesser than 2-methyl hexane

ii) n-butane knocks less than n-hexane.

The tendency to knock has been found to decrease in the order

n-alkane > mono substituted alkanes > cycloalkane > alkenes > Polysubstituted alkanes >aromatic compounds.

Octane Number:

Graham Edger proposed an arbitrary scale for expressing the knocking characteristics of gasoline in combustion engines as octane number in 1972 on the basis of antiknock properties. It has been found that n-heptane, CH_3-CH_2-CH_2-CH_2-CH_2-CH_2- CH_3, knocks very badly and hence, its antiknock value has been arbitrarily given zero. On the other hand, isooctane (2, 2, 4 – trimethylpentane) gives very little knocking, so its antiknock value is given as 100.

Thus, Octane number (or rating) of a gasoline (or any other internal combustion engine fuel) is the percentage of isooctane in a mixture of isooctane and n-heptane, which matches the fuel under test in knocking characteristics.

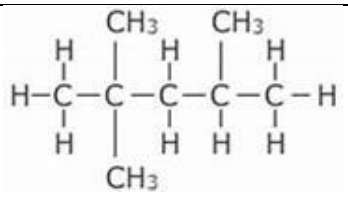
Reference Fuels	Chemical Structure	Octane
Isooctane (2,2,4 –trimethylpentane)		100
n-heptane	CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3	0

Table 1.2: Reference structures for Octane no scale

Ex: If a sample of petrol gives as much knocking as the mixture of 65% isooctane and 35% n-heptane, then the octane number of petrol is 65.

Anti-Knocking Agents

The octane rating of gasoline samples can be increased by the addition of certain organometallic compounds called anti-knocking agents and the process is called “doping”.

- An extensively used anti-knocking agent is tetraethyl lead $Pb(C_2H_5)4$ 0.5ml/litre. (but it causes air pollution by Pb.)
- Addition of alkylate fraction to petrol i.e. branched chain alkane of higher mol. wt.
- Addition of benzene, toluene, xylene (BTX) to petrol.

- Addition of methyl t-butyl ether (MTBE) or ethyl t-butyl ether.
- Adjusting suitable compression ratio

Quality of Petrol can be Improved by Adding:

- Antioxidant to improve oxidation resistance.
- Octane no. improvers like MTBE.
- Freezing point depressant like ethanol, glycol to minimize ice formation in fuel lines in cold regions.
- Detergents like fatty amines to clean fuel lines and carburettors.

Knocking in Diesel Engines

It works on the principle of compression ignition. Air is drawn into the cylinder and compressed to a pressure of 3.5 X 10^{-3} kgm^{-2}. During compression cylinder gets heated around 350 °C. Then diesel is introduced in the form of spray which is spontaneously get ignited and burns producing pressure on the piston. It takes some time for the fuel to get ignited. The time interval between the fuel injection and ignition is called **ignition delay**. It will be milliseconds. The **shorter ignition delay** leads to burning of the fuel at the rate at which it is injected. Whereas **longer ignition delay** results in accumulation of the fuel in the engine and this causes an explosive combustion when ignited. As a result, the **diesel engine knocks**.

Cetane Number of Diesel

Cetane Number is a measurement of the combustion quality of diesel fuel during compression ignition. It is a significant expression of the quality of a diesel fuel. Cetane number is a measure of a fuel's ignition delay, the time period between the start of injection and the first identifiable pressure increases during combustion of the fuel. In a particular diesel engine, higher cetane fuels will have shorter ignition delay periods than lower cetane fuels.

The suitability of a diesel fuel is determined by its cetane value which is the percentage of n-hexadecane in a mixture of n- hexadecane and 2-methyl naphthalene, which has the same ignition characteristics as the diesel sample.

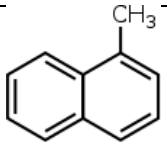
Reference Fuels	Chemical Structure	Cetane Number
α-methyl Naphthelene		0
Cetane (Hexadecane)	CH_3-$(CH_2)_{14}$-CH_3	100

Table 1.3: Reference structures for Cetane no scale

Relative order of cetane no. values

n-alkanes > cycloalkanes > alkenes > branched alkanes > Aromatics

How to increase Cetane No.

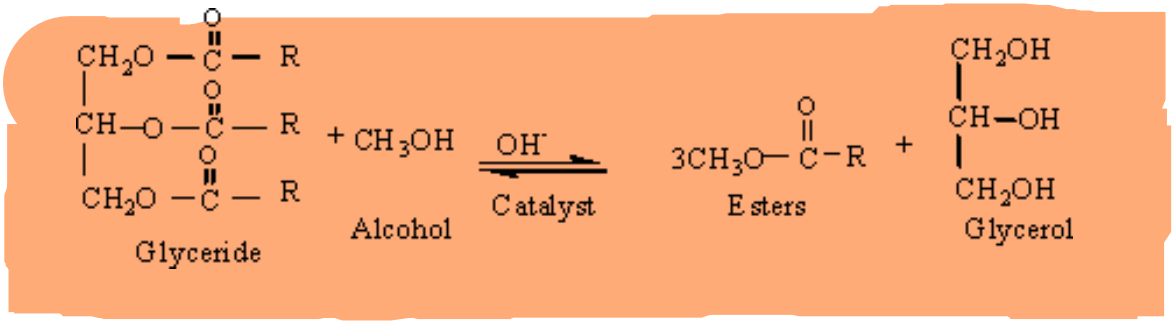
Alkyl nitrates (principally 2-ethylhexyl nitrate) and di-tert-butyl peroxide are used as additives to raise the cetane no.

- Antioxidant to improve oxidation resistance during storage.
- Lubricity additives for lubrication of fuel injection systems.
- Detergents to clean fuel line and carburetor.

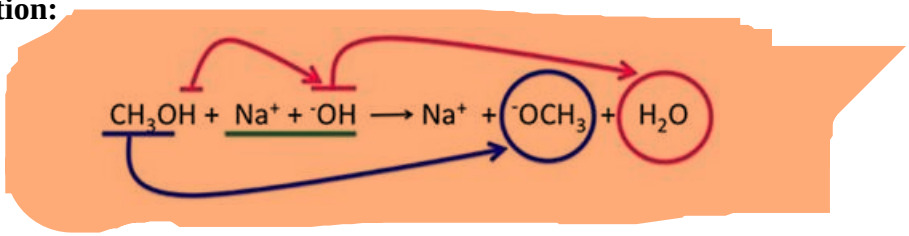
Biodiesel

Chemically biodiesel is the mixture of methyl esters of long chain carboxylic acids. It is eco-friendly fuel obtained from renewable sources like vegetable oils and animal fats. Common vegetable (edible and Non-edible) oils used are Soyabean oil, palm oil, peanut oil, cotton seed oil, corn oil, pongamial oil, Jatropha oil etc. Vegetable oils are the triglycerides, which have high viscosity, flash point and low heating value. Hence they cannot be used as such in diesel engines. Triglycerides need to be converted into biodiesel by a process called Transesterification. During the transesterification, the triglyceride is treated with methanol or ethanol in the presence of a base like KOH/NaOH, forms methyl or ethyl esters of fatty acids **(biodiesel)**

Conditions: Temperature: 55-60 C, Catalyst-KOH/NaOH, Reaction time-1-8 hrs.



Separation:



As the above two products are immiscible phases, we can easily drain off the glycerine/glycerol by washing with water and biodiesel can be collected.

Extra information:

Compounds present in biodiesel are like,

Methyl palmitate	$H_3C - (CH_2)_{14} - COOCH_3$
Methyl stearate	$H_3C - (CH_2)_{16} - COOCH_3$
Methyl oleate	$H_3C - (CH_2)_7 - CH = CH - (CH_2)_7 - COOCH_3$
Methyl linoleate	$H_3C - (CH_2)_5 - (CH = CH)_2 - (CH_2)_7 - COOCH_3$

Advantages of Biodiesel:

- Biodiesel is cheaper, as it is manufactured from cheap, non-edible or waste oil or animal fats.
- It has high cetane numbers 46 to 54 and high C.V of about 40 kJ/gm.
- It is regenerative and environment friendly.
- It does not give out particulate and CO pollutants, as Oxygen atoms in biodiesel help for complete combustion.
- It has certain extent of lubricity, due to higher oiliness of the esters.
- Its use provides good market to vegetable oils and reduces our dependence on diesel on foreign countries, saving currency.
- It is clean to use biodiesel in diesel engines.

Limitations of Biodiesel:

- Cloud and pour points of biodiesel are higher than diesel and can cause problems in the fuel flow line. So it cannot be used in cold regions.
- Biodiesel may have dissolving action rubber hoses, gaskets.
- There is a shortage of vegetable oils and the starting material if costly, the biodiesel will be costly.
- Biodiesel strongly adheres to metals and can become gummy.

Power Alcohol

When ethyl alcohol is used as fuel in an internal combustion engine, it is called power alcohol. It contains 25% of alcohol and 75% petrol.

Advantages of Power Alcohol:

- Ethyl alcohol has a good anti knocking property and its octane number is 90, while the octane number of petrol is about 65. Therefore, addition of ethyl alcohol to petrol increases its octane number.

- Alcohol has the property of absorbing any traces of water if present in petrol.
- If a specially designed engine with higher compression ratio is used, then the disadvantage of lower C.V. of ethyl alcohol can be overcome.
- Ethyl alcohol contains 'O' atoms, which help for complete combustion of power alcohol and the polluting emissions of CO, hydrocarbon, particulates are reduced largely.
- Use of ethyl alcohol in petrol reduces our dependence on foreign countries for petrol and saves foreign currency considerably.
- Power alcohol is cheaper than petrol.

Disadvantages of Power Alcohol:

- Ethyl alcohol has C.V. 7000 cal/gm much lower than C.V. of petrol 11500 cal/gm. Use of power alcohol reduces power output up to 35%.
- Ethyl alcohol has high surface tension and its atomization, especially at lower temperatures, is difficult causing starting trouble.
- Ethyl alcohol may undergo oxidation to form acetic acid, which corrodes engine parts.
- Ethyl alcohol obtained by fermentation directly cannot be mixed with petrol but it has to be dehydrated first.
- As ethyl alcohol contains 'O' atoms, the amount of air required for complete combustion of power alcohol is lesser and therefore the carburetor and engine need to be adjusted or modified, when only ethyl alcohol is used as fuel.

Green Fuels

Demand for green fuel technologies is rising given their potential to accelerate the transition to more sustainable forms of energy while still supporting current energy models with all their regional variations. Amongst all green fuels hydrogen is considered to be the best alternative. Hydrogen is a zero-emissions source of fuel for trains, buses and cars. It can be used as a feedstock gas for industries such as chemical, refining and steel. In addition, it is a source of heat and power for buildings, and can buffer energy generated from renewable sources.

Green fuel

Green fuel, also known as biofuel, is a type of fuel distilled from plants and animal materials.

OR

Green fuel is Carbon-neutral or even carbon-free alternatives to fossil fuels**.**

Types of Green fuels

(i) Biomass (ii) Geothermal energy (iii) Wind energy (iv) Solar energy (v) Hydroenergy

Why to split water?

- > 70% of Earth is made of water and which is abundantly available
- Emission of greenhouse gases can be reduced

Methods of water splitting

- Electrolysis
- Photoelectrochemical water splitting
- Photocatalytic water splitting
- Radiolysis
- Nuclear process
- Thermal process

Hydrogen Production

Over the last decades, an increasing interest has been devoted to the generation of hydrogen from solar-powered water splitting devices as a direct approach to generate a clean and sustainable energy vector. Among the different alternatives, a direct photocatalytic process is the most straightforward, where a light absorber is able to produce hydrogen and oxygen under light irradiation without an external bias.

Photocatalyst

Is a substance which functions as a catalyst when exposed to light.
Examples: TiO_2, ZnO, CdS, etc

Photo catalytic water splitting:

The photocatalytic generation of solar fuels by suitable photocatalytic materials suspended in water is certainly one of the most attractive and challenging pathways for decarbonizing current energy systems. From the outside, it seems as a simple approach combining water, solar light, and a semiconductor. However, each of these components plays a key role in the overall efficiency of the reaction, and mostly, many factors including thermodynamic, kinetic, and other requirements hold back this technology from its maximum potential. Photocatalysts must be semiconductors.

The minimum requirement is to generate electrons-hole pairs due to irradiation, normally by a UV-vis source. But it must also have a suitable crystalline phase, which ensures the formation of the electron-hole pairs and does not favor its recombination.

Material requirement of photo catalysts

(i) Band gap: Band gap of the materials used in catalysis should be more than 1.23 eV and sufficiently small to make efficient use of the solar spectrum (<3eV). And it should have band levels suitable for water splitting.

(ii) **High crystallinity:** Defects can act as recombination sites

(iii) **Long term stability:** Charge transfer used for water splitting and not for corrosion.

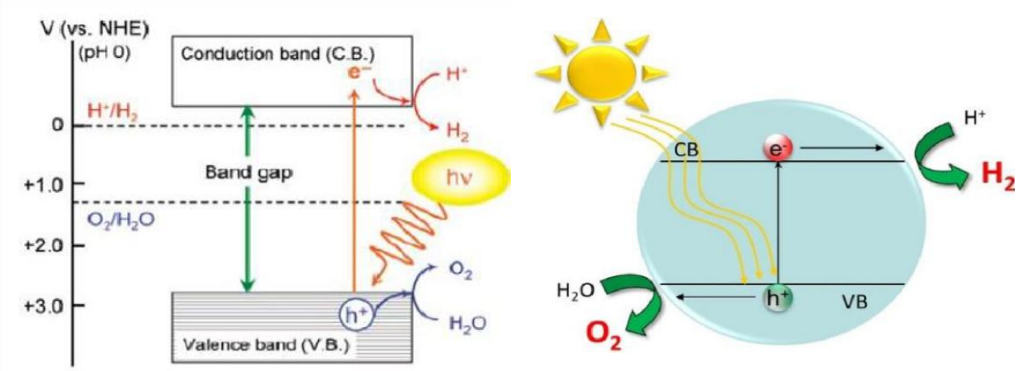


Figure 1.2: Band diagram of catalyst and Hydrogen production by water splitting

Water splitting electrode should be

(i) Efficient light absorber, (ii) charge separator (iii) charge conductor

Photocatalytic water splitting is an artificial photosynthesis process with photocatalysis in a photoelectrochemical cell used for the dissociation of water into its constituent parts, hydrogen and oxygen, using light. Water splitting electrodes work much like the leaves of the plants during photosynthesis. They absorb energy from the sunlight and use it to generate charge carriers. Negatively charged electrons and positively charged holes. These charges are delivered to water and water molecules swipe electrons away to form hydrogen gas and whereas holes strip the molecules of electrons to form oxygen gas.

Mechanism of water splitting

(i) When light incident on material, ie., photon with energy 1.23 eV (wavelength less than 1000 nm) is absorbed. it promotes electron from valence bond to conduction band.

(ii) Generation of electron hole pair in conduction and valence bond respectively and migrate to the surface.

(iii) Redox reaction taking place at the surface the material. Adsorbed species (water) is reduced and oxidized by the electrons and holes.

$2H_2O \rightarrow 2H_2 + O_2$

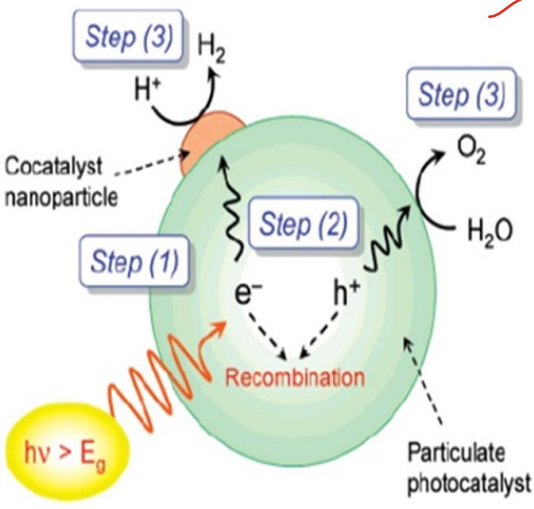


Figure 1.3: Photo catalytic water splitting steps

Hydrogen Storage

Hydrogen energy storage is an area of growing interest due to the greater storage capacity offered by hydrogen over batteries, as well as the benefits offered by the renewable generation afforded by hydrogen as a fuel. Hydrogen can be stored either as a gas or as a liquid. Hydrogen gas storage typically requires the use of high-pressure tanks (350-700 bar or 5000-10,000 psi), while liquid hydrogen storage requires cryogenic temperatures to prevent it boiling back into a gas (which occurs at −252.8°C). Hydrogen can also be stored on the surface of (adsorption) or within solid materials (absorption).

General method of hydrogen storage

In modern practice hydrogen can be stored in three different ways

1. As a gas under high pressures (in pressure cylinder)
2. In liquid form under cryogenic temperatures (at low temperature storage)
3. On the surface of or within solid and liquid materials (Metal hydrides as well as carbon nanotubes and metal organic frameworks)

Rocket Propellant:

Rocket propellants are the fuels and the oxidizers carried by the rocket for propulsion. Propellants are heterogeneous mixtures of fuels and small particulate oxidizers, which are held together by a rubbery material referred to as the binder.

- The propellant is a material that spews out of the back of the spacecraft giving it thrust, or a push forward.
- Often the propellant is a kind of fuel which is burned with an oxidizer to produce large volumes of very hot gas. These gasses expand until they rush out of the back of the rocket, making thrust.
- A fuel propellant is often burned with an oxidizer propellant to produce large volumes of very hot gas.

Chemical rockets create thrust by reacting propellants within a combustion chamber into a very hot gas at high pressure, which is then expanded and accelerated by passage through a nozzle at the rear of the rocket.

Examples: Ammonium perchlorate (AP) as oxidizer and hydrazine as a fuel are today widely used as propellants.

Oxidizers: Ammonium perchlorate (AP); other oxidizers are metal perchlorates, ammonium nitrate (AN), and ammonium dinitramide (ADN) etc

Fuels: Hydrazine, monomethyl hydrazine (MMH), and unsymmetric dimethylhydrazine (UDMH).

Characteristics:

Rocket propellants possess unique properties, such as

- A capability to self-sustain the burning process.
- Generate thermal energy, and simultaneously produce propulsive mass.
- Able to self-ignite (initiate burning without outside power input).
- Can burn in vacuum.
- Low cost and availability in large quantities. The "specific gravity" is the ratio of the density of the propellant to that of water. A higher specific gravity corresponds to a denser propellant. The advantage of denser propellants is that they require less storage.
- Space for a given amount of mass. Less storage space can have favorable consequences for a vehicle.
- Propellant stability, to be distinguished from combustion stability to be discussed later, refers to the ability of a propellant to maintain quantity and composition under various conditions without degradation. "Long-term stability" refers to the ability of a propellant to maintain quantity and composition during storage over long periods of time (10–15 years).
- Should have Short ignition delay The ignition delay (determined by a combination of the propellant chemical properties and the manner in which the propellants are introduced into the combustion chamber) should be reasonably short to prevent unburned propellants from building up in the combustion chamber and then exploding.

This is because a propellant consists of two integral components: a fuel that burns and produces propulsive mass and an oxidizer that facilitates and sustains oxidation. In this respect, rocket propellants are more like explosives rather than like automobile and aviation fuels which require atmospheric air for oxidation.

The major difference between explosives and rocket propellants is the **gas expansion rate** which is much slower in propellants and makes it possible to contain and control the process of burning inside the rocket engine. The gases produced when a fuel propellant is burned, expand and push on a nozzle, which accelerates them until they rush out of the back of the rocket at extremely high speed, making thrust.

There are three main types of propellants: solid, liquid, and hybrid.

(1) Solid propellants consist of a solid oxidizer and a solid fuel. Solid propellants for space applications are today widely used in large boosters for launchers and, to some extent, for in-space propulsion. Propellants for these applications are based on the oxidizer ammonium perchlorate (AP), NH_4ClO_4, and aluminum powder embedded in a polymer binder matrix.

Classification with examples:

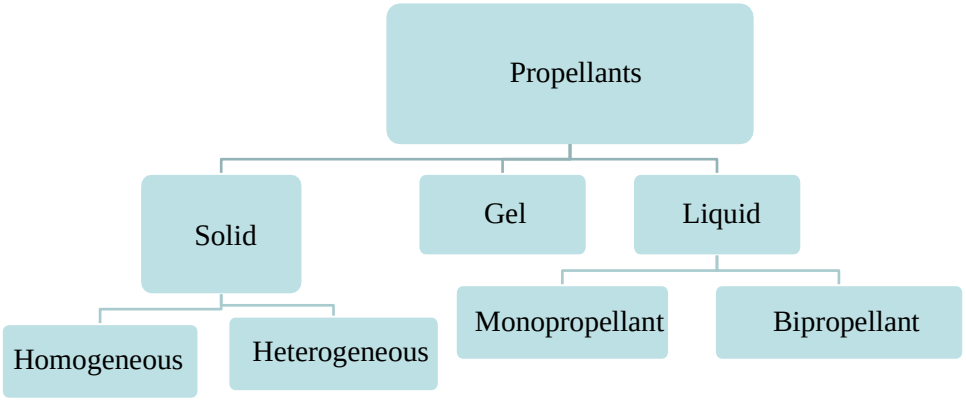


Figure 1.4: Classification of chemical propellants

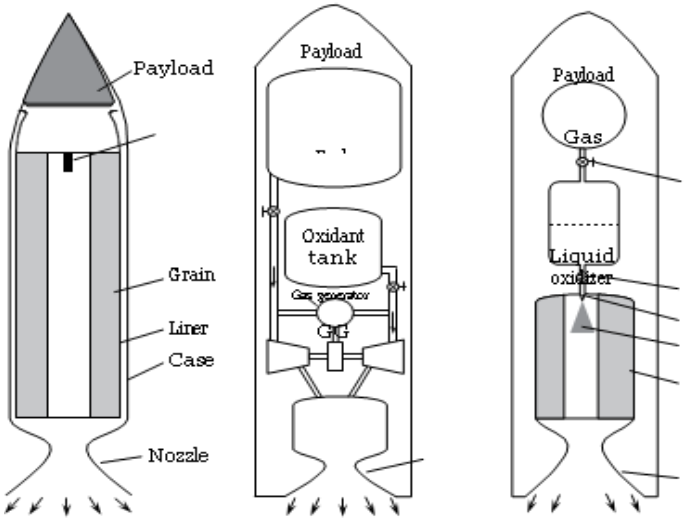


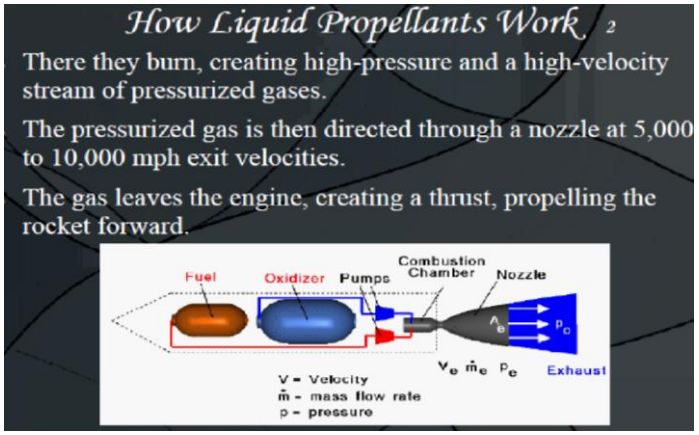
Figure 1.5: Three types of chemical rocket engines: (a) solid propellant, liquid propellant, and (c) hybrid propellant

In the case of gunpowder, the fuel is charcoal, the oxidizer is potassium nitrate, and sulfur serves as a catalyst. Ammonium Perchlorate Composite Propellant (APCP), Nitrogen tetroxide (N_2O_4) and hydrazine (N_2H_4). Solid-fueled rockets are much easier to store and handle than liquid-fueled rockets, which makes them ideal for military applications.

Liquid propellants:

Liquid fueled rockets have higher specific impulse than solid rockets and are capable of being throttled, shut down, and restarted. Only the combustion chamber of a liquid fueled rocket needs to withstand combustion pressures and temperatures and they can be regeneratively cooled by the liquid propellant. On vehicles employing turbo pumps, the propellant tanks are at very much less pressure than the combustion chamber, and thus can be built far more lightly than a solid propellant rocket case, permitting a higher mass ratio. For these reasons, most

orbital launch vehicles use liquid propellants. The primary performance advantage of liquid propellants is due to the oxidizer.



Several practical liquid oxidizers (liquid oxygen, nitrogen tetroxide, and hydrogen peroxide) are available which have much better specific impulse than the ammonium perchlorate used in most solid rockets, when paired with comparable fuels. **Liquid Fuels:** Hydrogen, Kerosene, Ethyl alcohol

(1) Hybrid propellants:

Hybrid propellants are those propellants, which are a mixture of solid and liquid propellants. In these propellants, one of the two components (oxidizer and fuel) is solid (usually fuel) whereas the other is liquid (usually oxidizer). In a hybrid propellant rocket engine, the liquid part is injected into the solid part. Thus the storage chamber of the solid part acts as the combustion chamber. In a hybrid rocket the oxidizer flows down the perforation (see solid propellants) in the solid fuel grain and reacts with the solid fuel. This produces the hot exhaust gases required to produce thrust. This process can be seen in the following image: In many hybrid motor designs, the oxidizer is pressurized **liquefied nitrous oxide (N_2O)** while the fuel is **cellulose ($C_6H_{10}O_5$)**.

Applications of rocket engines

In comparison to other air-breathing engines, rocket engines have very specific applications, which can be broadly divided into four types: (1) space launch vehicle, (2) spacecraft, (3) missile, and (4) miscellaneous. Rocket engines have been used for several other civilian applications depending on the imagination of people. Rocket engines that are used routinely for collecting weather predictions are commonly known as “sounding rockets.” Rocket engines are also used for research in airplanes, in rocket-assist take off, and for providing lifelines to ships under distress. Besides this, they are considered for use in developing propulsion belts. Rockets can be used in the future for providing relief materials to inaccessible places during natural calamities.